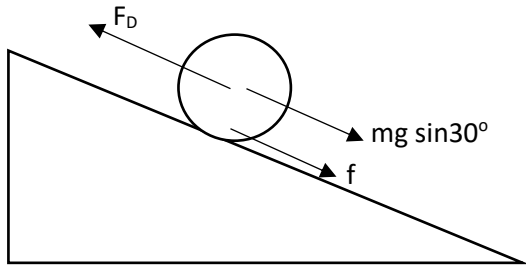


The graph of  $WD$  against  $x$  should have a decreasing gradient.

6

Ans: C



Find driving force  $F_D$ :

$$F_D - f - mg \sin 30^\circ = ma$$

$$F_D - 900 - mg \sin 30^\circ = 1200(3.0)$$

$$F_D = 10386 \text{ N}$$

Find power delivered by engine:

$$P = F_D v$$

$$P = (10386)(20)$$

$$P = 208 \text{ kW}$$